AISC 5020 Mathematics & Statistics for AI

Lecture 1 | Introduction

Gary Choi (CUHK Mathematics)

9 September 2026

Course information

- Course Code: AISC 5020
- Course Title: Mathematics & Statistics for AI
- Teaching Mode: On-site face-to-face
- Unit: 3 Credits
- Medium of Instruction: English
- Time: Wednesday, 18:30 - 21:15
- Venue: Lecture Theatre 3, G/F, Lee Shau Kee Building (LSK)

This course provides a solid foundation in the **mathematical and statistical concepts** necessary for understanding and developing AI models, including linear algebra, calculus, optimization, probability, and statistical inference.

Teaching team

Teaching staff:

- (Course Convenor) Dr. CHEUNG King Chau Simon (kingchaucheung@cuhk.edu.hk), Department of Statistics and Data Science
- Prof. CHOI Pui Tung Gary (ptchoi@cuhk.edu.hk), Department of Mathematics

Teaching assistants:

- DANISH Muhammad (mdanish@link.cuhk.edu.hk), Department of Statistics and Data Science
- PAN Tse Fung Kingston (Kingstonpan@link.cuhk.edu.hk), Department of Statistics and Data Science
- HOU Liguang (lghou@link.cuhk.edu.hk/lghou@math.cuhk.edu.hk), Department of Math
- LAU Kai Kwan (kaikwan.lau@link.cuhk.edu.hk/kklau@math.cuhk.edu.hk), Department of Math
- LAU Trenton (trentonlau@link.cuhk.edu.hk/ttlau@math.cuhk.edu.hk), Department of Math

Learning outcomes

Gain the **mathematical and statistical foundations** needed to understand, design, and build AI models.

- Use linear algebra (vectors, matrices, eigenvalues, SVD) to structure data and representations
- Apply multivariable calculus and computation graphs to derive and implement backpropagation
- Utilize probability, information theory, and inference to formulate principled objectives
- Diagnose conditioning and stability, choose effective SGD/optimizer strategies, and reason about constraints, reduce dimensionality with PCA, model latent structure via GMM/EM, leverage the mathematics of CNNs for vision...

Altogether, a **disciplined ability** to translate real problems into well-optimized AI systems.

Course Schedule

Lecture	Date	Topic	Course Instructor (Unit)
1	9 September 2026	**Introduction** - AI pipeline: data -> model -> loss -> optimization - Why math is essential in AI - Overview of regression, classification, generative modelling - Quick review: sets, vectors, probability language	Prof. CHOI Pui Tung Gary (PMA)
2	16 September 2026	**Linear Algebra I** - Scalars, vectors, matrices and tensors - Dot products, cosine similarity - L1/L2 norms, distances, orthogonality - Applications: similarity, embeddings	Prof. CHOI Pui Tung Gary (PMA)

Course Schedule

3	23 September 2026	**Linear Algebra II** - Linear independence, rank, inverses, span - Eigen-decomposition, SVD - Conditioning of matrices - Applications: Solving system of linear equations and SVD based solvers for stable computation, a binary classification of nodes in a graph via the algebraic connectivity of its Laplacian, the importance of eigenvectors in analysing linear systems.	Prof. CHOI Pui Tung Gary (PMA)
4	30 September 2026	**Vector Calculus and Computation Graphs** - Derivatives, gradients, Jacobian, Hessians - Multivariate chain rule - Taylor series - Computation graphs - Automatic differentiation	Prof. CHOI Pui Tung Gary (PMA)

Course Schedule

5	7 October 2026	**Probability Foundations** - Random variables, PMFs/PDFs, expectations, variance - Joint, marginal, conditional distributions - Covariance matrices - Common distributions: Gaussian, Bernoulli, Multinomial - Independence and conditional independence	Dr CHEUNG King Chau Simon (STA)
6	14 October 2026	**Information Theory: Entropy and Cross-Entropy and KL Divergence** - Entropy, cross-entropy as loss - KL divergence as distribution discrepancy - Why CE = H + KL and why CE is used for classification	Dr CHEUNG King Chau Simon (STA)

Course Schedule

7	21 October 2026	**Statistical Inference** - Likelihood and loglikelihood - Least squares and MLE - MSE loss and its gradient - MAP estimation and regularization	Dr CHEUNG King Chau Simon (STA)
8	28 October 2026	**Bayesian Statistics and Bayesian Networks** - Bayes' rule, priors, posteriors, conjugate families - Bayesian networks: structure, inference, factorization, conditional independence - DAGs, d-separation, local marginals	Dr CHEUNG King Chau Simon (STA)

Course Schedule

9	4 November 2026 (Midterm Test)	**Midterm Test for the first 1.5 – 2 hours** **Optimization I: Fundamentals** - Gradient-based optimization - Convexity, convex functions, convex sets - Line search, step size, condition number, stability	Prof. CHOI Pui Tung Gary (PMA)
10	11 November 2026	**Optimization II: Constrained Optimizations** - Lagrange multipliers (equality constraints) - Karush-Kuhn-Tucker Theorem, inequality constraints - Duality theorem - Problem formulation - Complementary slackness - Saddle-point interpretation - Application: Quadratic program	Prof. CHOI Pui Tung Gary (PMA)

Course Schedule

11	18 November 2026	**Optimization III: Stochastic Gradient Descent (SGD)** - Stochastic Gradient Descent - Mini-batch SGD, momentum, RMSProp, Adam - Convergence issues in high dimensions - Practical concerns: learning rate schedules, initialization	Prof. CHOI Pui Tung Gary (PMA)
12	25 November 2026	**Dimensionality Reduction** - PCA via eigenvectors; SVD formulation - Low-rank approximation - Probabilistic PCA (link to latent variable modelling)	Dr CHEUNG King Chau Simon (STA)

Course Schedule

13	2 December 2026	**Probabilistic Generative Models** - Gaussian Mixture Models (GMMs) - EM algorithm - Generative vs Discriminant models - Probabilistic Generative Models (GANs, VAEs)	Dr CHEUNG King Chau Simon (STA)
14	9 December 2026	**Final Exam**	Dr CHEUNG King Chau Simon (STA)

Assessment

Assessment Type(s)	Percentage (100%)
Homework Assignments (4 assignments in total)	30%
Midterm Test (4 November 2026, in-class)	20%
Group Project	25%
Final Exam (9 December 2026)	25%

Assessment

Assessment Type(s)	Percentage (100%)
Homework Assignments (4 assignments in total)	30%
Midterm Test (4 November 2026, in-class)	20%
Group Project	25%
Final Exam (9 December 2026)	25%

For the homework assignments:

- HW1: covering the content of Lectures 1-4
- HW2: covering the content of Lectures 5-8
- HW3: covering the content of Lectures 9-11
- HW4: covering the content of Lectures 12-13

Assessment

Assessment Type(s)	Percentage (100%)
Homework Assignments (4 assignments in total)	30%
Midterm Test (4 November 2026, in-class)	20%
Group Project	25%
Final Exam (9 December 2026)	25%

For the group project, please:

- check the *Group Project Guidelines* and *Dr. Cheung's announcement* on Blackboard
- form your groups as early as possible (typically 4 to 6 students per group)
- email us the list of group members by September 13, 2026 (check *TA Kingston's message* on Blackboard)

Students who have not joined a group by the deadline will be **randomly grouped** with other students who are also without a group.

Academic Honesty and Plagiarism

- The University places **very high importance on honesty in academic work** submitted by students, and adopts a policy of zero tolerance on academic dishonesty.
- Any related offence will lead to disciplinary action including termination of studies at the University. Everyone should make himself/herself familiar with the content of this website and thereby help avoid any practice that would not be acceptable.
- Details may be found at

  `http://www.cuhk.edu.hk/policy/academichonesty/`
- With each assignment, students will be required to submit a signed declaration that they are aware of these policies, regulations, guidelines and procedures.
- See the **Course Outline PDF on Blackboard** for details.

Use of AI Tools

Approach 3 - Use of AI tools is allowed with explicit acknowledgement and proper citation

- Students may use some AI tools in some learning activities and/or assessments on the condition that they make explicit acknowledgement and proper citations of the input from AI tools.
- The use of any AI tools is strictly prohibited in all forms of mid-term tests, quizzes, and examinations.
- Acknowledging support from AI tools: Students are required to acknowledge all functional uses of an AI tool and cite it when they paraphrase, quote, or incorporate into their own work any content (whether it is text, image, data, or other format) that was created by it.
- Improper/unauthorized use of AI tools in learning activities and assessments will constitute acts of academic dishonesty which will be handled in accordance with the University's Procedures for Handling Cases of Academic Dishonesty.
- See the **Course Outline PDF on Blackboard** for details.

Required Readings

- Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep learning. Cambridge, MA: MIT Press.
- Deisenroth, M. P., Faisal, A. A., & Ong, C. S. (2020). Mathematics for machine learning. Cambridge, England: Cambridge University Press.
- Nelson, H. (2023). Essential math for AI: Next-level mathematics for developing efficient and successful AI systems. Sebastopol, CA: O'Reilly Media.

Recommended Readings

- Trevor Hastie, Robert Tibshirani, Jerome Friedman (2001). The Elements of Statistical Learning.
- Chris Bishop (2006). Pattern Recognition and Machine Learning.
- Luc Devroye, LaszloGyorfi, Gabor Lugosi. (1996). A probabilistic theory of pattern recognition.
- Gyorfi, Kohler, Krzyzak and Walk (2002). A Distribution-Free Theory of Nonparametric Regression.
- Larry Wasserman (2004). All of Statistics: A Concise Course in Statistical Inference.
- Aggarwal, C. C. (2020). Linear algebra and optimization for machine learning: A textbook. Cham, Switzerland: Springer Nature Switzerland AG.
- Golden, R. M. (2020). Statistical machine learning: A unified framework. CRC Press.

AI pipeline

An AI pipeline is typically as follows:



Repeat on training data; evaluate on held-out data.

A useful AI model needs:

- a suitable representation
- a meaningful error measure
- a way to improve its parameters

Why math is essential in AI?

An AI pipeline is typically as follows:



Repeat on training data; evaluate on held-out data.

Question	**Mathematical tools**
How do we represent measurements?	Vectors and matrices
How does an output change?	Calculus
What is uncertain?	Probability and statistics
How do we improve a model?	Optimization

What's next for this lecture?

Brief introduction to:

- Regression
- Classification
- Generative modelling

Review of some basic mathematical concepts:

- Basic arithmetic and functions
- Set theory
- Probability
- Vectors and matrices
- Calculus

Three broad tasks in AI for science

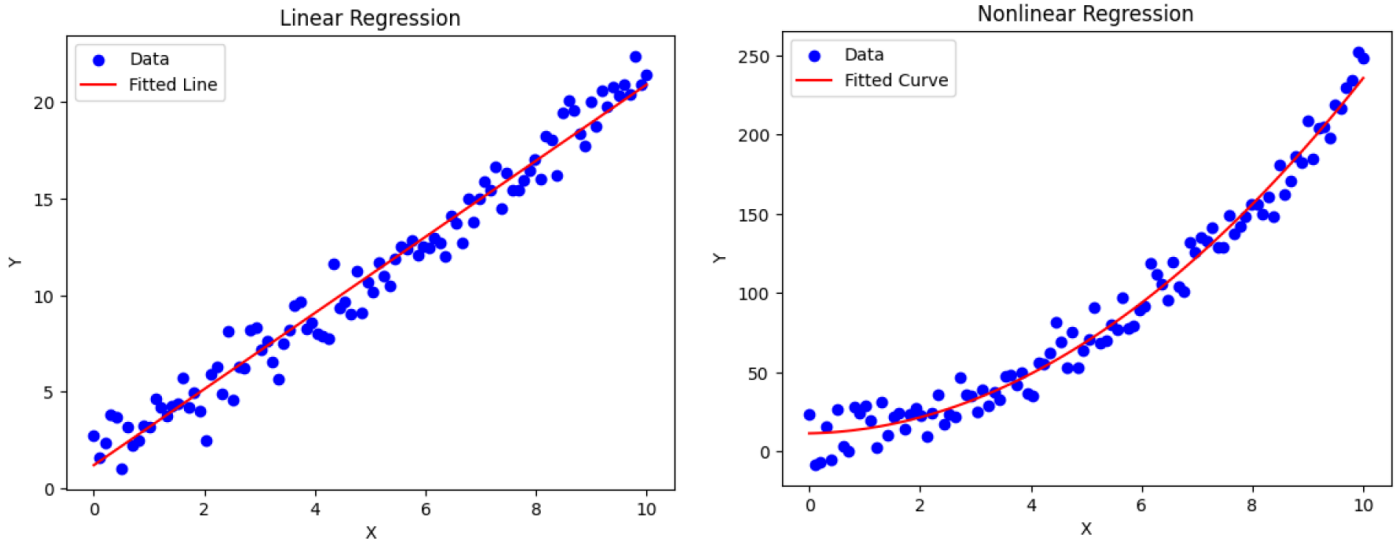
Task	Desired output	Example
Regression	Number	Predicted concentration
Classification	Category or class probability	Predicted material type
Generative modelling	New structured sample	Candidate molecular structure

Regression

How can we construct a model based on known data to make reasonable predictions?

Applications:

- Science, Medicine, Engineering, Business, ...



(Image source: GeeksforGeeks)

Classification

How can we construct a model that can assign data points to predefined categories/classes/labels?
Applications:

- Biology, Biometrics, Pattern recognition, Search engines, ...

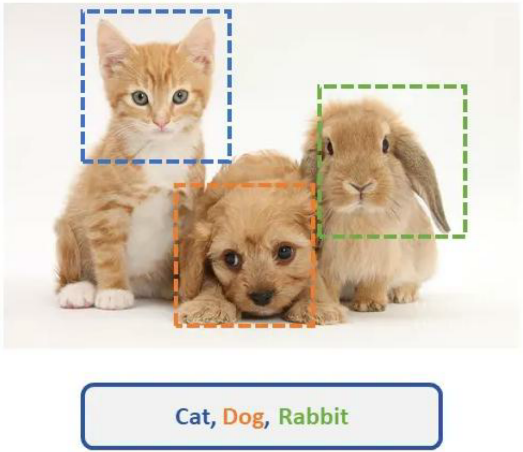


Image source: Beyond Entropy

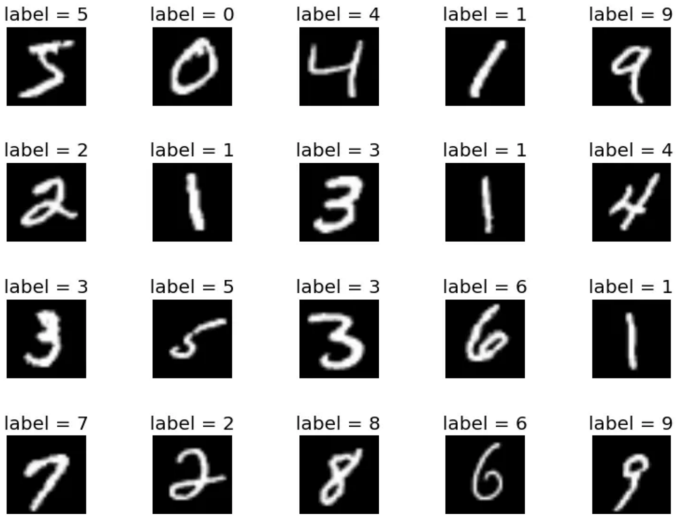


Image source: MNIST dataset

Example: The MNIST database

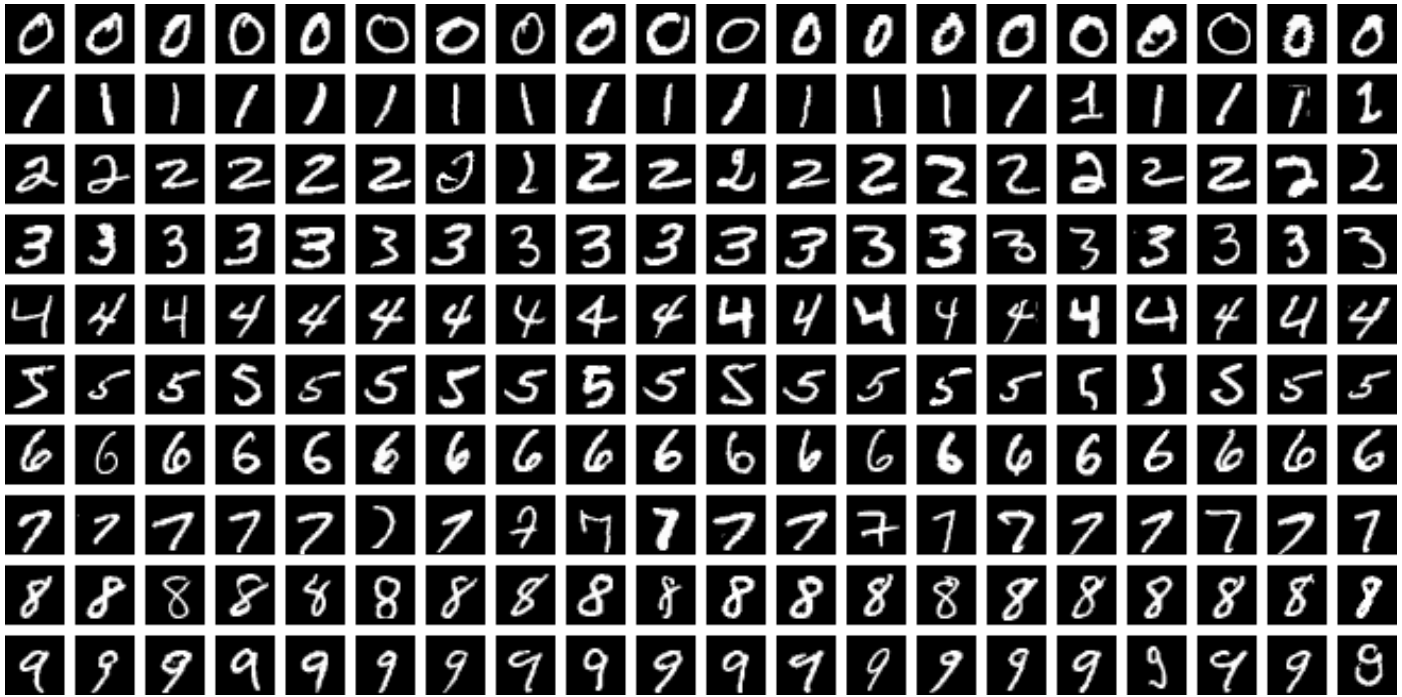
MNIST: Modified National Institute of Standards and Technology database

- 70,000 images of handwritten digits ($0 - 9$) with 28×28 pixel grayscale image
- Taken from some American Census Bureau employees and high school students
- Each image has an associated truth label indicating the digit it represents
- A widely used benchmark dataset



Example: The MNIST database

More examples:



Regression vs classification

Both regression and classification are about prediction, but there are some notable differences between them!

Regression:

- Predict a numeric value
- Focus more on least-squares models
- Goal: Finding the underlying trend for predicting numerical outcomes

Classification:

- Predict a label/class/category
- May consider more abstract and sophisticated models and loss functions
- Goal: Finding decision boundaries that separate classes

Regression vs classification

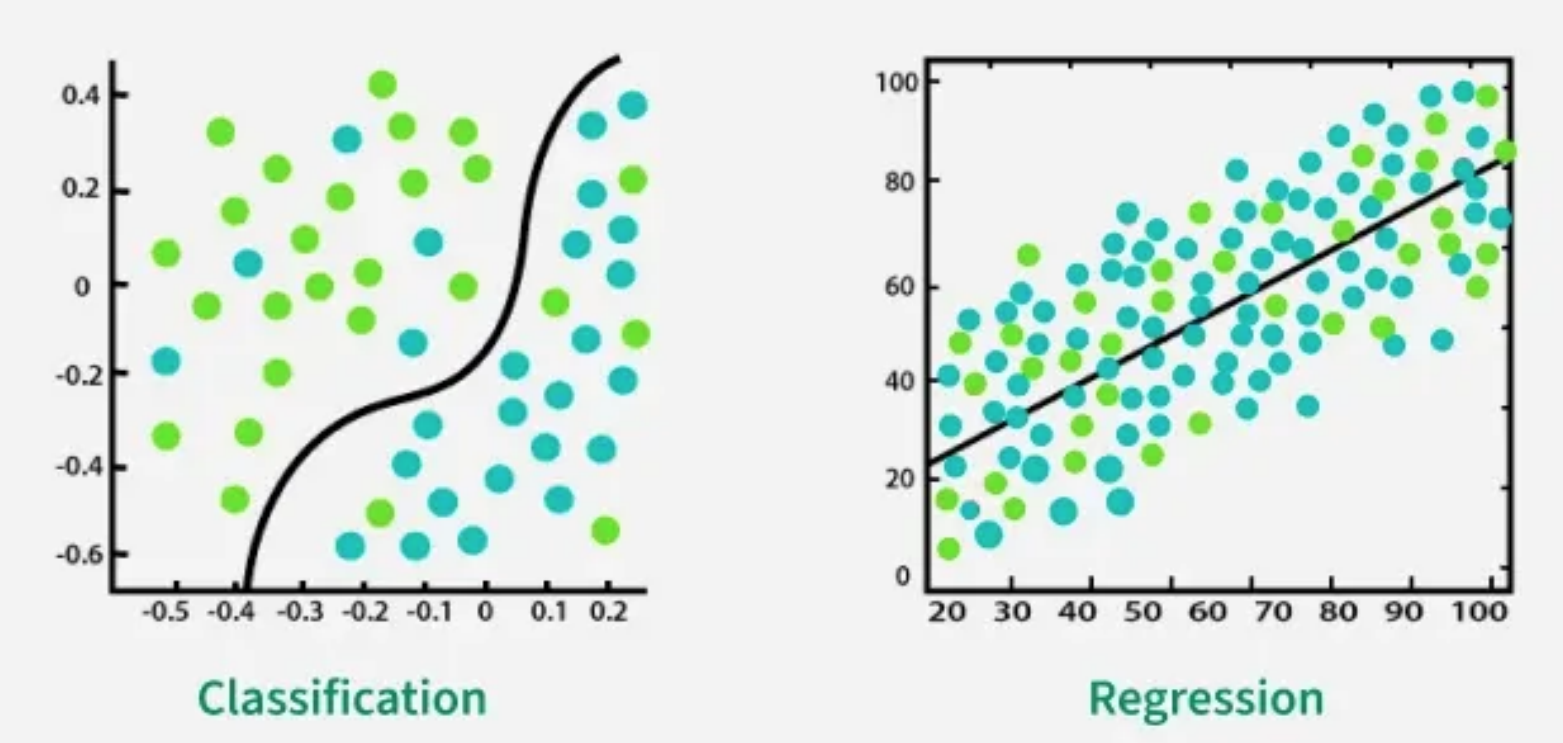
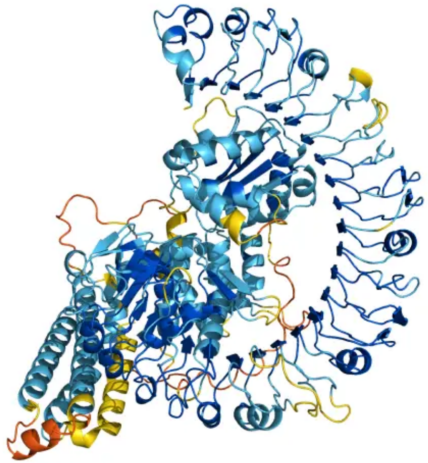


Image source: GeeksforGeeks

Generative modelling

How can we learn underlying distributions of complex scientific data to simulate, design, and discover novel physical, chemical, and biological system?
Applications:

- Chemistry and Drug Discovery, Materials Science, Cosmology & Physics...



(Source: AlphaFold)

Three broad tasks in AI for science

Task	Desired output	Example
Regression	Number	Predicted concentration
Classification	Category or class probability	Predicted material type
Generative modelling	New structured sample	Candidate molecular structure

The **scientific question** determines what task is to be performed.

Example: For a collection of chemical samples:

- Predict reaction yield: regression
- Predict whether yield exceeds a chosen threshold: classification
- Propose a new candidate sample: generative modelling

The same raw data may be used for different tasks.

Three broad tasks in AI for science

Task	Desired output	Example
Regression	Number	Predicted concentration
Classification	Category or class probability	Predicted material type
Generative modelling	New structured sample	Candidate molecular structure

Question: For each of the following scientific problems, what is the suitable task and what are the inputs/outputs?

- Estimate tomorrow's river level from recent sensor readings.
- Label a microscopy patch as containing a cell or not.
- Propose new simulated microstructures with specified porosity.

Three broad tasks in AI for science

Task	Desired output	Example
Regression	Number	Predicted concentration
Classification	Category or class probability	Predicted material type
Generative modelling	New structured sample	Candidate molecular structure

Question: For each of the following problems, what is the suitable task and what are the inputs/outputs?

- Estimate tomorrow's river level from recent sensor readings.
  Regression; input recent readings, output a level.
- Label a microscopy patch as containing a cell or not.
  Classification; input a patch, output a class probability or label.
- Propose new simulated microstructures with specified porosity.
  Generative modelling; input desired porosity and randomness, output a structure.

Three broad tasks in AI for science

Task	Desired output	Example
Regression	Number	Predicted concentration
Classification	Category or class probability	Predicted material type
Generative modelling	New structured sample	Candidate molecular structure

More generally, how do we achieve the three tasks above using mathematics and statistics?

Function

A **function** turns an input into an output.
Example: $f(x) = x^2 + 1$ we have $f(0) = 0^2 + 1 = 1, \quad f(2) = 2^2 + 1 = 5, \ldots$
More precisely, we may write $f : X \to Y$ where X is the **domain** that contains allowed inputs, and Y is the **codomain** that specifies the output type.
Some common functions:

- Linear function: $f(x) = 2x + 3$
- Quadratic function: $f(x) = x^2 + 4x - 1$
- Trigonometric function: $f(x) = 2\sin(3x)$
- Exponential function: $f(x) = e^x = exp(x)$ (where $e = 2.71828\ldots$ is the Euler's number)
- Logarithmic function: $\log(x)$, $\ln(x)$ (base-e logarithm)

Remarks. Functions can also take multiple numbers as input:

$$f(x, y) = x^2 + y + 2, \quad f(1, 0) = 1^2 + 0 + 2 = 3$$

Later, we will see that functions can also take **vectors** or **matrices** as input, and we will use **set theory** to better describe the domain and codomain.

Sequences of numbers, summations, and products

We usually use letters with subscripts, e.g., $x_1, x_2, x_3, x_4, \ldots$ to represent a sequence of numbers.
Examples:

- $a_n = \frac{1}{n} = 1, \frac{1}{2}, \frac{1}{3}, \ldots$
- $b_n = 2^{n-1} = 1, 2, 4, 8, \ldots$
- $c_n = (-1)^n = -1, 1, -1, 1, \ldots$
- Arithmetic sequences: $a_{n+1} - a_n = d$ for some constant d
- Geometric sequences: $a_{n+1} = ra_n$ for some constant r

Remarks. Later, we will see that data points are usually also represented using subscripts, such as (x_1, y_1), (x_2, y_2), ...

Sequences of numbers, summations, and products

The **summation** of a sequence of numbers is usually denoted by the Σ symbol:

$$\sum_{i=1}^{n} x_i = x_1 + x_2 + x_3 + \cdots + x_n$$

Example. If $x_1 = 2,\ x_2 = 4,\ x_3 = 6,\ x_4 = 8$, then the sum is

$$\sum_{i=1}^{4} x_i = x_1 + x_2 + x_3 + x_4 = 2 + 4 + 6 + 8 = 20$$

The index i is a dummy variable that represents the sequentially changing integer values.

The **product** of a sequence of numbers is usually denoted by the Π symbol:

$$\prod_{i=1}^{n} p_i = p_1 \times p_2 \times p_3 \times \cdots \times p_n.$$

Example. For $p_1 = \frac{1}{2}, p_2 = \frac{1}{4}, p_3 = \frac{1}{2}$, the product is

$$\prod_{i=1}^{3} p_i = p_1 \times p_2 \times p_3 = \frac{1}{2} \times \frac{1}{4} \times \frac{1}{2} = \frac{1}{16}.$$

Set: a collection of objects

- We use curly brackets $\{...\}$ to define a **set**, which is a collection of objects.
  Example:
  $$S = \{a, b, c\} = \text{a set containing three elements } a, b, c$$

- $x \in S$ means "x is an element of the set S"
- $x \notin S$ means "x is NOT an element of the set S"
- $|S|$: the number of distinct objects in the set S.

  Example: For $S = \{5, 0, 2, 0\}$, we have
  - $2 \in S$
  - $1 \notin S$
  - $|S| = 3$

Set notations

- $A \subset B$ (also written as $A \subseteq B$) means "A is a subset of B"
  - i.e., for any element $x \in A$, we have $x \in B$
- We can also define a set by specifying the properties of the elements it collects. The usual notations are:

$$\{x : \cdots\}, \quad \{x | \cdots\}, \quad \{x \text{ such that } \cdots\}$$

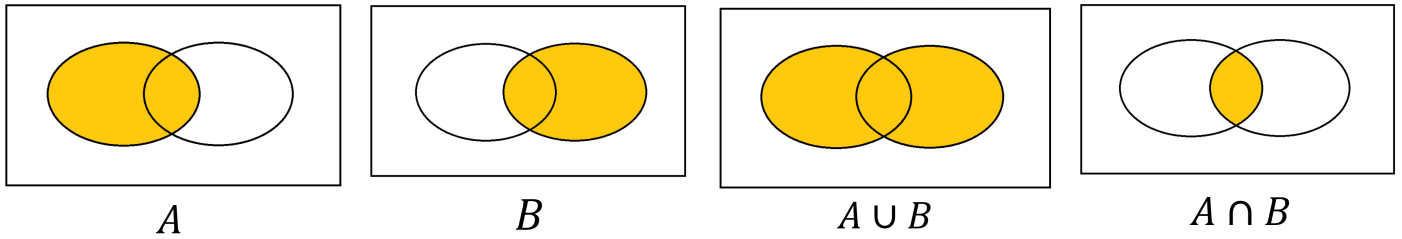
- $\mathbb{R}$ = the set of all real numbers
- $\mathbb{Z} = \{\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\}$ = the set of all integers
- $\mathbb{N} = \mathbb{Z}^+ = \{x \in \mathbb{Z} : x > 0\} = \{1, 2, 3, \ldots\}$ = the set of all positive integers
- $\mathbb{Q} = \{x \in \mathbb{R} : x = \frac{p}{q} \text{ for some } p, q \in \mathbb{Z} \text{ with } q \neq 0\}$ = the set of all rational numbers
- $\emptyset = \{\ \}$ = empty set

Examples:

- $2 \in \mathbb{Z}$ (since 2 is an integer), $\pi \notin \mathbb{Q}$ (since π is an irrational number), $\{0, 2, 4, 6, \ldots\} \subset \mathbb{Z}$

Set intersections and unions

- Union of two sets: $A \cup B = \{x : x \in A \text{ or } x \in B\}$
- Intersection of two sets: $A \cap B = \{x : x \in A \text{ and } x \in B\}$
- Union of multiple sets $A_1, A_2, \ldots, A_n$: $\bigcup_{i=1}^{n} A_i = A_1 \cup A_2 \cup \cdots \cup A_n$
- Intersection of multiple sets $A_1, A_2, \ldots, A_n$: $\bigcap_{i=1}^{n} A_i = A_1 \cap A_2 \cap \cdots \cap A_n$
- Set difference: $A \setminus B = \{x : x \in A \text{ and } x \notin B\}$



Set intersections and unions

- Union of two sets: $A \cup B = \{x : x \in A \text{ or } x \in B\}$
- Intersection of two sets: $A \cap B = \{x : x \in A \text{ and } x \in B\}$
- Union of multiple sets $A_1, A_2, \ldots, A_n$: $\bigcup_{i=1}^{n} A_i = A_1 \cup A_2 \cup \cdots \cup A_n$
- Intersection of multiple sets $A_1, A_2, \ldots, A_n$: $\bigcap_{i=1}^{n} A_i = A_1 \cap A_2 \cap \cdots \cap A_n$
- Set difference: $A \setminus B = \{x : x \in A \text{ and } x \notin B\}$

Examples:

- $\{1, 2, 3\} \cup \{1, 3, 4, 7\} = \{1, 2, 3, 4, 7\}$
- $\{1, 2, 3\} \cap \{1, 3, 4, 7\} = \{1, 3\}$
- $\{1, 2, 3\} \setminus \{1, 3, 4, 7\} = \{2\}$

Set intersections and unions

Intervals:

- $(a, b) = \{x \in \mathbb{R} : a < x < b\}$ (open interval)
- $[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}$ (closed interval)
- $(a, b] = \{x \in \mathbb{R} : a < x \leq b\}$
- $[a, b) = \{x \in \mathbb{R} : a \leq x < b\}$
- $(a, \infty) = \{x \in \mathbb{R} : x > a\}$
- $[a, \infty) = \{x \in \mathbb{R} : x \geq a\}$
- $(-\infty, b) = \{x \in \mathbb{R} : x < b\}$
- $(-\infty, b] = \{x \in \mathbb{R} : x \leq b\}$

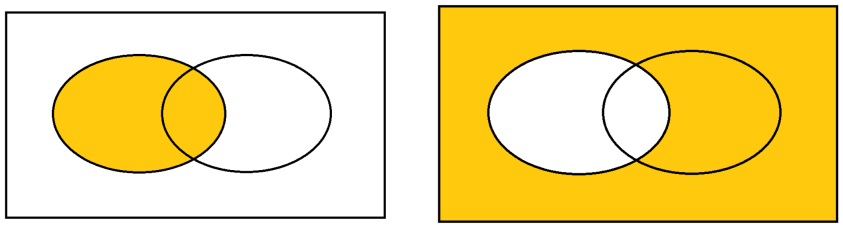
Examples:

- $(-1, 3) \cup (0, 4] = (-1, 4]$
- $[0, 5] \cap (1, \infty) = (1, 5]$
- $(0, 5) \setminus (1, 2) = (0, 1] \cup [2, 5)$
- $\bigcup_{n \in \mathbb{Z}} [2n\pi, (2n+1)\pi) = \cdots \cup [-2\pi, -\pi) \cup [0, \pi) \cup [2\pi, 3\pi) \cup \cdots$

Complement of set

The **complement** of a set A is the collection of all elements in the universal set Ω that do not belong to A. It is usually denoted by A' or A^c. In other words, we have

$$A^c = \Omega \setminus A = \{x \in \Omega : x \notin A\}$$



Example. Given the universal set (the set of all samples) $\Omega = \{1, 2, 3, 4, 5\}$ and a set $A = \{1, 2, 4\}$, the complement of A is

$$A^c = \Omega \setminus A = \{3, 5\}.$$

Example

Let $\Omega = \{1, 2, 3, 4, 5, 6\}$, $A = \{1, 3, 5\}$ and $B = \{3, 4, 5\}$.

- Find $A \cap B$, $A \cup B$ and A^c.
- Is $|A \cup B| = |A| + |B|$?

Example

Let $\Omega = \{1, 2, 3, 4, 5, 6\}$, $A = \{1, 3, 5\}$ and $B = \{3, 4, 5\}$.

- Find $A \cap B$, $A \cup B$ and A^c.
- Is $|A \cup B| = |A| + |B|$?

Answer:

$$A \cap B = \{3, 5\}, \quad A \cup B = \{1, 3, 4, 5\}, \quad A^c = \{2, 4, 6\}.$$

We have $|A \cup B| = 4$ but $|A| + |B| = 3 + 3 = 6$, so they are not equal.

More generally, we have the following **Inclusion–exclusion principle**:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

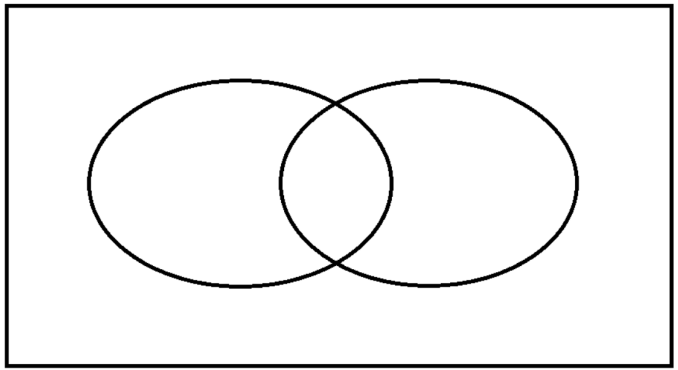
Sets

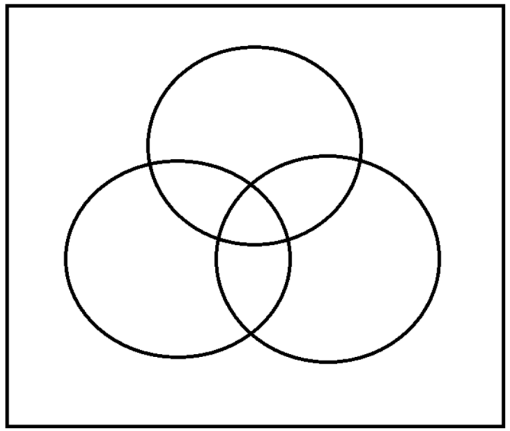
- **Theorem** (De Morgan's Law)

$$(A \cap B)' = A' \cup B'$$
$$(A \cup B)' = A' \cap B'$$

- **Theorem** (Distributive Laws)

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$
$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$





Probability

Probability gives a numerical measure for the degree of uncertainty or the occurrence of an event.

If there are n total possible outcomes in a sample space S, and m of those are favorable for an event A, then the probability of event A is given as

$$P(A) = \frac{\text{number of favorable outcomes}}{\text{total number of possible outcomes}} = \frac{n(A)}{n(S)} = \frac{m}{n}$$

Example. Find the probability of getting a 3 or 5 when throwing a dice.

Solution. Sample space $S = \{1, 2, 3, 4, 5, 6\}$ and event $A = \{3, 5\}$.

We have $n(A) = 2$ and $n(S) = 6$. So, $P(A) = \frac{n(A)}{n(S)} = \frac{1}{3}$.

Set and Probability

We can also use the set language to describe the events in probability:

- The **complement** of event A is the set of all outcomes in a sample that are not included in the event A. The complement of event A is denoted by A'.
- The event $A \cap B$ is the **intersection** of the events A and B and consists of outcomes that are contained within both events A and B.
- Two events are said to be **mutually exclusive** if $A \cap B = \emptyset$.
- The event $A \cup B$ is the **union** of events A and B and consists of the outcomes that are contained within at least one of the events A and B.

Laws of probability

- **Definition** (Axioms of Probability) For an experiment with sample space $S = \{e_1, e_2, \cdots, e_n\}$, we can assign probabilities $P(e_1), P(e_2), \cdots, P(e_n)$ provided that:
  - $0 \leq P(e_i) \leq 1$
  - $P(e_1) + P(e_2) + \cdots + P(e_n) = 1$.

- **Theorem** (Complement Rule)
$$P(A') = 1 - P(A)$$

- **Theorem** (Addition Law) If A and B are two different events then
$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Counting rules

- The number of permutations of n distinct objects taken r at a time is

$$ {}_nP_r = \frac{n!}{(n-r)!} $$

- The number of distinct subsets or combinations of size r that can be selected from n distinct objects, $(r \leq n)$, is given by

$$ {}_nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!} $$

Conditional probability and independence

- The **conditional probability** of event A given B is

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \text{ for } P(B) > 0$$

- The events A and B are called **independent** if

$$P(A \cap B) = P(A)P(B)$$

- Events $B_1, B_2, \cdots, B_k$ are said to be a **partition** of the sample space S if the following two conditions are satisfied:
  (1) $B_i \cap B_j = \emptyset$ for each pair i, j
  (2) $B_1 \cup B_2 \cup \cdots \cup B_k = S$

Conditional probability and independence

Example. Suppose 2% of a group of people have influenza, and the remaining 98% are healthy:

$$P(\text{sick}) = 2\% = 0.02, \quad P(\text{healthy}) = 98\% = 0.98,$$

When testing an individual, there may be some errors:

	Positive	Negative
Sick	True positive $P(\text{positive} \mid \text{sick}) = 95\%$	False negative $P(\text{negative} \mid \text{sick}) = 5\%$
Healthy	False positive $P(\text{positive} \mid \text{healthy}) = 3\%$	True negative $P(\text{negative} \mid \text{healthy}) = 97\%$

Then:

- Probability of getting a healthy individual who is tested negative
  $= P(\text{negative} \cap \text{healthy}) = P(\text{healthy}) \times P(\text{negative}|\text{healthy}) = 0.98 \times 0.97 = 0.9506$
- Probability of getting a sick individual who is tested positive
  $= P(\text{positive} \cap \text{sick}) = P(\text{sick}) \times P(\text{positive}|\text{sick}) = 0.02 \times 0.95 = 0.019$

Examples

The probability that John passes a Math exam is $\frac{4}{5}$ and that he passes a Chemistry exam is $\frac{5}{6}$. If the probability that he passes both exams is $\frac{3}{4}$, find the probability that he will pass at least one exam.

Examples

The probability that John passes a Math exam is $\frac{4}{5}$ and that he passes a Chemistry exam is $\frac{5}{6}$. If the probability that he passes both exams is $\frac{3}{4}$, find the probability that he will pass at least one exam.

Answer:
Let $M =$ John passes the math exam, and $C =$ John passes the chemistry exam.
$P(\text{John passes at least one exam}) = P(M \cup C) = P(M) + P(C) - P(M \cap C) = \frac{4}{5} + \frac{5}{6} - \frac{3}{4} = \frac{53}{60}$

Examples

A package of 6 light bulbs contains 2 defective bulbs. If 3 bulbs are selected for use, find the probability that none of the three is defective.

Examples

A package of 6 light bulbs contains 2 defective bulbs. If 3 bulbs are selected for use, find the probability that none of the three is defective.

Answer:

$$P(\text{none are defective})$$
$$= \frac{\text{number of ways 3 non defectives can be chosen}}{\text{total number of ways a sample of 3 can be chosen}}$$
$$= \frac{\binom{4}{3}}{\binom{6}{3}} = \frac{1}{5}$$

Examples

Three bits (0 or 1 digits) are transmitted over a noisy channel, so they will be flipped independently with probability 0.1 each. What is the probability that

1. At least one bit is flipped?
2. Exactly one bit is flipped?

Examples

Three bits (0 or 1 digits) are transmitted over a noisy channel, so they will be flipped independently with probability 0.1 each. What is the probability that

1. At least one bit is flipped?
2. Exactly one bit is flipped?

Answer:

1. Using the complement rule, $P(\text{at least one}) = 1 - P(\text{none})$. If we denote F_k the event that the kth bit is flipped, due to independence, we have

$$\begin{aligned} P(\text{at least one}) &= 1 - P(\text{no bits are flipped}) \\ &= 1 - P(F_1' \cap F_2' \cap F_3') \\ &= 1 - (1 - 0.1)^3 = 0.271 \end{aligned}$$

2. Flipping exactly one bit can be accomplished in 3 ways:

$$P(\text{exactly one}) = P(F_1 \cap F_2' \cap F_3') + P(F_1' \cap F_2 \cap F_3') + P(F_1' \cap F_2' \cap F_3) = 0.243$$

Bayes theorem

For two dependent random events A and B with $P(A) > 0$ and $P(B) > 0$, then the probability $P(A|B)$ and probability $P(B|A)$ are related by:

$$P(A|B) = \frac{P(A)P(B|A)}{P(B)}$$

Example. Let A be the event that a student chooses CUHK for postgraduate studies, and B be the event that a student chooses our AISCI programme. Given Probabilities:

- $P(A) = 0.80$
- $P(B) = 0.16$
- $P(B|A) = 0.09$

Then we have:

$$P(A|B) = \frac{P(A)P(B|A)}{P(B)} = \frac{0.8 \times 0.09}{0.16} = 0.45$$

Discrete Probability Distributions

- A **random variable (RV)** is a number associated with each outcome of some random experiment. A random variable X is said to be **discrete** if it can take on only a finite or countable number of possible values x.
  Example. Toss two coins and record the number of heads: 0, 1, or 2. Then, the following outcomes can be observed.

Outcome	TT	HT	TH	HH
Number of heads	0	1	1	2

- The random variables will be denoted by capital letters $X, Y, Z, \cdots$, and the lowercase x will represent a particular value of X. For the example above, $x = 2$ if heads comes up twice. For the probability that the random variable X has the value x, we write $P(X = x)$, or just $p(x)$.
  Example. For the coin flipping random variable X, we can make the table:

x	0	1	2
$p(x)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

  The table represents the **probability distribution** of the random variable X.

Probability Mass Function

The function $p_X(x)$ or simply $p(x)$ is called the **Probability mass function (PMF)** of X if it satisfies:

- $P(X = x) = p_X(x) \geq 0$
- $\sum_x P(X = x) = 1$, where the sum is over all possible x

Example. A shipment of 8 computers contains 3 that are defective. If a school makes a random purchase of 2 of these computers, find the PMF for the number of defectives.

Solution. Let X be a random variable whose values x are the possible numbers of defective computers purchased by the school. Then x must be $0, 1$ or 2. For each case, we have

$$P(X = 0) = \frac{\binom{3}{0}\binom{5}{2}}{\binom{8}{2}} = \frac{10}{28}, \quad P(X = 1) = \frac{\binom{3}{1}\binom{5}{1}}{\binom{8}{2}} = \frac{15}{28}, \quad P(X = 2) = \frac{\binom{3}{2}\binom{5}{0}}{\binom{8}{2}} = \frac{3}{28}$$

Thus, the PMF of X is given by

x	0	1	2
$p(x)$	$\frac{10}{28}$	$\frac{15}{28}$	$\frac{3}{28}$

Cumulative Distribution Function

The **cumulative distribution function (CDF)** $F(x)$ for a random variable X is defined as

$$F(x) = P(X \leq x)$$

If X is discrete, $F(x) = \sum_{y \leq x} p(y)$ where $p(x)$ is the probability mass function.

Example. Find the CDF of the random variable in the last example.

Solution. The CDF of the random variable X is:

$F(0) = p(0) = \frac{10}{28}$, $\quad F(1) = p(0) + p(1) = \frac{10}{28} + \frac{15}{28} = \frac{25}{28}$,

$F(2) = p(0) + p(1) + p(2) = \frac{10}{28} + \frac{15}{28} + \frac{3}{28} = 1$

Hence,

$$F(x) = \begin{cases} 0 & \text{for } x < 0 \\ \frac{10}{28} & \text{for } 0 \leq x < 1 \\ \frac{25}{28} & \text{for } 1 \leq x < 2 \\ 1 & \text{for } x \geq 2 \end{cases}$$

Expected value

The **mean** or **expected value** of a discrete random variable X with probability mass function $p(x)$ is given by

$$\mathbb{E}(X) = \sum_x xp(x)$$

The **variance** of a random variable X with expected value μ is given by

$$V(X) = \sigma^2 = \mathbb{E}(X - \mu)^2 = \mathbb{E}(X^2) - \mu^2,$$

where

$$\mathbb{E}(X^2) = \sum_x x^2 p(x)$$

The **standard deviation** of a random variable X is the square root of the variance, and is given by

$$\sigma = \sqrt{V(X)} = \sqrt{\mathbb{E}(X - \mu)^2}$$

Remark. The mean describes the *center* of the probability distribution, while the standard deviation describes the *spread*.

Expected value

Example. The number of fire emergencies at a rural county in a week has the following distribution:

x	0	1	2	3	4
$P(X = x)$	0.52	0.28	0.14	0.04	0.02

Find $\mathbb{E}(X)$, $V(X)$ and σ.

Solution. By definition, we see that

$$\mathbb{E}(X) = 0(0.52) + 1(0.28) + 2(0.14) + 3(0.04) + 4(0.02) = 0.76 = \mu$$

$$\mathbb{E}(X^2) = 0^2(0.52) + 1^2(0.28) + 2^2(0.14) + 3^2(0.04) + 4^2(0.02) = 1.52$$

$$V(X) = 1.52 - (0.76)^2 = 0.9424$$

$$\sigma = \sqrt{0.9424} \approx 0.9708$$

Vectors

An object of the form $(a_1, a_2, \cdots, a_n)$ is called an **(n-dimensional) vector**.

- The elements $a_1, a_2, \cdots, a_n$ are called the **entries**, or **components** of the vector.
- More specifically, **row vectors** take the form $(a_1, a_2, \cdots, a_n)$ and **column vectors** take the form $\begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}$

Examples:

- Below are some examples of row vectors: $\begin{pmatrix} 1 & 3 \end{pmatrix}$, $\begin{pmatrix} 2 & -1 & 0 & 10 & 7.5 \end{pmatrix}$
- Below are some examples of column vectors: $\begin{pmatrix} 1 \\ 3 \end{pmatrix}$, $\begin{pmatrix} 3 \\ -1 \\ 4 \\ -1.59 \end{pmatrix}$

Vectors

Remarks.

- The set of all n-dimensional vectors such that $a_1, a_2, \cdots, a_n$ are real numbers is denoted by $\mathbb{R}^n$. e.g.$(3, -2, 0)$ and $(-1, 1, 4)$ are both in $\mathbb{R}^3$.
- Vectors can be used for keeping several measurements together, e.g.,

$$x = \begin{bmatrix} 20 \\ 7 \\ 0.4 \end{bmatrix} \in \mathbb{R}^3 \quad \longleftrightarrow \quad \begin{bmatrix} \text{temperature} \\ \text{pH} \\ \text{signal} \end{bmatrix}$$

- However, note that the order of the entries is important and cannot be freely changed

Matrices

An $m \times n$ **matrix** with entries from $\mathbb{R}$ is a rectangular array of the form

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix},$$

where each entry a_{ij} $(1 \leq i \leq m, 1 \leq j \leq n)$ is a real number.

- We call the entries a_{ij} with $i = j$ the **diagonal entries** of the matrix.
- The entries $a_{i1}, a_{i2}, \cdots, a_{in}$ compose the ***i*th row** of the matrix, and the entries $a_{1j}, a_{2j}, \cdots, a_{mj}$ compose the ***j*th column** of the matrix.
- We usually denote the entry of a matrix A that lies in row i and column j by A_{ij}.
- If the number of rows and columns of a matrix are equal, we call it a **square matrix**.

Example. Some examples of matrices: $\begin{pmatrix} 1 & 3 \end{pmatrix}$, $\begin{pmatrix} 2 & 1 \\ 3 & 8 \end{pmatrix}$, $\begin{pmatrix} 1 & 3 & 5 \\ 7 & 9 & 11 \end{pmatrix}$, $\begin{pmatrix} 2 & 7 \\ 1 & 8 \\ 2 & 8 \end{pmatrix}$

Vector addition and scalar multiplication

We can do coordinate-wise addition and scalar multiplication for vectors. More specifically, if $u = (a_1, a_2, \cdots, a_n) \in \mathbb{R}^n$, $v = (b_1, b_2, \cdots, b_n) \in \mathbb{R}^n$, and $c \in \mathbb{R}$, then

$$u + v = (a_1 + b_1, a_2 + b_2, \cdots, a_n + b_n)$$

$$cu = (ca_1, ca_2, \cdots, ca_n)$$

Example. In $\mathbb{R}^3$, we have

$$(3, -2, 0) + (-1, 1, 4) = (2, -1, 4)$$

$$-5(1, -2, 0) = (-5, 10, 0)$$

Example

Calculate $-2u + 4v$ where $u = \begin{pmatrix} -2 \\ -2 \\ 2 \end{pmatrix}$ and $v = \begin{pmatrix} 0 \\ 4 \\ 5 \end{pmatrix}$.

Example

Calculate $-2u + 4v$ where $u = \begin{pmatrix} -2 \\ -2 \\ 2 \end{pmatrix}$ and $v = \begin{pmatrix} 0 \\ 4 \\ 5 \end{pmatrix}$.

Answer:

$$-2u + 4v = -2\begin{pmatrix} -2 \\ -2 \\ 2 \end{pmatrix} + 4\begin{pmatrix} 0 \\ 4 \\ 5 \end{pmatrix} = \begin{pmatrix} 4 \\ 4 \\ -4 \end{pmatrix} + \begin{pmatrix} 0 \\ 16 \\ 20 \end{pmatrix} = \begin{pmatrix} 4 \\ 20 \\ 16 \end{pmatrix}$$

Matrix addition and scalar multiplication

We can also do coordinate-wise addition and scalar multiplication for matrices. More specifically, if $A, B \in M_{m\times n}(\mathbb{R})$ (where $M_{m\times n}(\mathbb{R})$ denotes the set of all $m \times n$ matrices with entries from $\mathbb{R}$) and $c \in \mathbb{R}$, we have

$$(A + B)_{ij} = A_{ij} + B_{ij}$$

$$(cA)_{ij} = cA_{ij}$$

for $1 \leq i \leq m$ and $1 \leq j \leq n$.

Example.

$$\begin{pmatrix} 2 & 0 & -1 \\ 1 & -3 & 4 \end{pmatrix} + \begin{pmatrix} -5 & -2 & 6 \\ 3 & 4 & -1 \end{pmatrix} = \begin{pmatrix} -3 & -2 & 5 \\ 4 & 1 & 3 \end{pmatrix}$$

and

$$-3\begin{pmatrix} 1 & 0 & 2 \\ -3 & 2 & 3 \end{pmatrix} = \begin{pmatrix} -3 & 0 & -6 \\ 9 & -6 & -9 \end{pmatrix}$$

Example

Calculate

1. $\begin{pmatrix} 2 & 5 & -3 \\ 1 & 0 & 7 \end{pmatrix} + \begin{pmatrix} 4 & -2 & 5 \\ -5 & 3 & 2 \end{pmatrix}$
2. $4 \begin{pmatrix} 2 & 5 & 3 \\ 1 & 0 & 7 \end{pmatrix}$

Example

Calculate

1. $\begin{pmatrix} 2 & 5 & -3 \\ 1 & 0 & 7 \end{pmatrix} + \begin{pmatrix} 4 & -2 & 5 \\ -5 & 3 & 2 \end{pmatrix}$
2. $4 \begin{pmatrix} 2 & 5 & 3 \\ 1 & 0 & 7 \end{pmatrix}$

Answer:

1. $\begin{pmatrix} 6 & 3 & 2 \\ -4 & 3 & 9 \end{pmatrix}$
2. $\begin{pmatrix} 8 & 20 & -12 \\ 4 & 0 & 28 \end{pmatrix}$

Matrix multiplication

Let A be an $m \times n$ matrix and B be an $n \times p$ matrix. We define the **product** of A and B, denoted as AB, to be the $m \times p$ matrix such that

$$(AB)_{ij} = A_{i1}B_{1j} + A_{i2}B_{2j} + \cdots + A_{in}B_{nj} = \sum_{k=1}^{n} A_{ik}B_{kj} \quad \text{for } 1 \leq i \leq m, 1 \leq j \leq p$$

Remark. In order for the product AB to exist, there are restrictions regarding the relative sizes of A and B. The following mnemonic device is helpful: $(m \times n) \cdot (n \times p) = (m \times p)$. That is, the two "inner" dimensions must be equal, and the two "outer" dimensions yield the size of the product.

Example. We have

$$\begin{pmatrix} 1 & 2 & 1 \\ 0 & 4 & -1 \end{pmatrix} \begin{pmatrix} 4 \\ 2 \\ 5 \end{pmatrix} = \begin{pmatrix} 1 \cdot 4 + 2 \cdot 2 + 1 \cdot 5 \\ 0 \cdot 4 + 4 \cdot 2 + (-1) \cdot 5 \end{pmatrix} = \begin{pmatrix} 13 \\ 3 \end{pmatrix}$$

Note that the symbolic relationship $(2 \times 3) \cdot (3 \times 1) = (2 \times 1)$.

Matrix multiplication

Matrix-vector multiplication $A\mathbf{x}$

$=$ a linear combination of columns of A

$$\begin{bmatrix} 2 & 1 & 3 \\ 3 & 1 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2x_1 + x_2 + 3x_3 \\ 3x_1 + x_2 + 4x_3 \end{bmatrix} = x_1 \begin{bmatrix} 2 \\ 3 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + x_3 \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

- $A : m \times n, \quad \mathbf{x} : n \times 1, \quad A\mathbf{x} : m \times 1$

Matrix-matrix multiplication $AB = C$

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 0 & 1 & 2 & 3 \\ 4 & 5 & 6 & 7 \\ 8 & 9 & 10 & 11 \end{bmatrix} = \begin{bmatrix} 32 & 38 & 44 & 50 \\ 68 & 83 & 98 & 113 \end{bmatrix}$$

- $A : m \times n, \quad B : n \times p, \quad C : m \times p$

Some special matrices

The $n \times n$ **identity matrix** I_n is defined by $(I_n)_{ij} = 1$ if $i = j$ and $(I_n)_{ij} = 0$ if $i \neq j$.

Example.

$$I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

We can easily verify that $AI_n = I_n A$ for all $n \times n$ matrices A.

A **diagonal matrix** is a matrix with non-zero entries only appearing in the diagonal entries, i.e.,

$$A = \begin{bmatrix} \times & & \\ & \ddots & \\ & & \times \end{bmatrix}$$

Examples.

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \quad \begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 5 \end{pmatrix},$$

Some special matrices

Upper triangular matrix: a square matrix where all the values below the main diagonal are zero

$$A = \begin{bmatrix} \times & \times & \dots & \times \\ 0 & \times & \dots & \times \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \dots & 0 & \times \end{bmatrix}$$

Lower triangular matrix: a square matrix where all the values above the main diagonal are zero

$$A = \begin{bmatrix} \times & 0 & \dots & 0 \\ \times & \times & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ \times & \dots & \times & \times \end{bmatrix}$$

Example. $A = \begin{pmatrix} 2 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 5 \end{pmatrix}$ is upper triangular, and $B = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 5 \end{pmatrix}$ is lower triangular

Matrix transpose

The **transpose** A^T of an $m \times n$ matrix A is the $n \times m$ matrix obtained from A by interchanging the rows with the columns; that is,

$$(A^T)_{ij} = A_{ji}$$

Example.

$$\begin{pmatrix} 1 & -2 & 3 \\ 0 & 5 & -1 \end{pmatrix}^T = \begin{pmatrix} 1 & 0 \\ -2 & 5 \\ 3 & -1 \end{pmatrix}.$$

We say that a matrix A is a **symmetric matrix** if $A^T = A$

We have the following properties for the transpose of vector and matrix products:

$$(A\mathbf{x})^T = \mathbf{x}^T A^T,\ (AB)^T = B^T A^T, (ABC)^T = C^T B^T A^T$$

Matrix inverse

Let A be an $n \times n$ matrix. Then A is **invertible** if there exists an $n \times n$ matrix B such that

$$AB = BA = I$$

If A is invertible, then the matrix B such that $AB = BA = I$ is unique. The matrix B is called the **inverse** of A and is denoted by A^{-1}.

For any 2×2 matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, it can be shown that the matrix inverse is

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Example. The inverse of $\begin{pmatrix} 5 & 7 \\ 2 & 3 \end{pmatrix}$ is $\begin{pmatrix} 3 & -7 \\ -2 & 5 \end{pmatrix}$.

Determinant (2×2 matrices)

If $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is a 2×2 matrix with entries from $\mathbb{R}$, then we define the **determinant** of A, denoted by $\det(A)$ or $|A|$, to be: $\det(A) = ad - bc$

Example: For the matrices $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 3 & 2 \\ 6 & 4 \end{pmatrix}$, we have

$$\det(A) = 1 \cdot 4 - 2 \cdot 3 = -2 \quad \text{and} \quad \det(B) = 3 \cdot 4 - 2 \cdot 6 = 0$$

Determinant (for general square matrices)

The **determinant** of square matrices is defined using three rules:

1. $\det(I) = 1$
2. Swapping two rows of the matrix $\Rightarrow$ det flips sign
3. Linearity in single-row operations
   - Multiplying a row by a scalar $\alpha \Rightarrow$ the determinant multiplies by α

$$\det \begin{bmatrix} \vec{a}_1 \\ \vdots \\ \alpha\vec{u} \\ \vdots \\ \vec{a}_k \end{bmatrix} = \alpha \det \begin{bmatrix} \vec{a}_1 \\ \vdots \\ \vec{u} \\ \vdots \\ \vec{a}_k \end{bmatrix}$$

   - Row addition rule:

$$\det \begin{bmatrix} \vec{a}_1 \\ \vdots \\ \vec{u}+\vec{v} \\ \vdots \\ \vec{a}_k \end{bmatrix} = \det \begin{bmatrix} \vec{a}_1 \\ \vdots \\ \vec{u} \\ \vdots \\ \vec{a}_k \end{bmatrix} + \det \begin{bmatrix} \vec{a}_1 \\ \vdots \\ \vec{v} \\ \vdots \\ \vec{a}_k \end{bmatrix}$$

Some useful properties of determinant

- $\det(A) \neq 0$ if and only if A is invertible
- $\det(\alpha A) = \alpha^n \det(A)$ (where A is an $n \times n$ matrix)
- $\det(AB) = \det(A)\det(B)$
- $\det(A^{-1}) = \frac{1}{\det(A)}$ (if A^{-1} exists)
- $\det(A^T) = \det(A)$
- $\det(\text{triangular matrix}) = \text{product of all diagonal entries}$

Some useful properties of determinant

- (For 2×2 matrices) $\det\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = ad - bc$
- (For 3×3 matrices) $\det\left(\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}\right) = a\det\left(\begin{bmatrix} e & f \\ h & i \end{bmatrix}\right) - b\det\left(\begin{bmatrix} d & f \\ g & i \end{bmatrix}\right) + c\det\left(\begin{bmatrix} d & e \\ g & h \end{bmatrix}\right)$
- A contains two equal rows (or columns) $\Longrightarrow \det(A) = 0$
- Adding a scalar multiple of any row (or column) to any row (or column) does not change the determinant
- General strategy for finding $\det(A)$: Apply row (or column) swaps and eliminations until we get det(triangular matrix)

Solving linear systems

A system of linear equations can be written as

$$A\mathbf{x} = \mathbf{b}$$

where A: $m \times n$ matrix, $\mathbf{x}$: $n \times 1$ vector, $\mathbf{b}$: $m \times 1$ vector

- $m = n$: **Square** linear system
  - Same number of equations (rows) and unknowns (columns)
  - Unique solution if all rows/columns are linearly independent (to be covered next time)
- $m > n$: **Overdetermined** system
  - More equations than unknowns
  - No true solution, can only solve in a least-squares sense
- $m < n$: **Underdetermined** system
  - More unknowns than equations
  - Multiple solutions, can choose the "best" solution based on some extra criterion

Vector norm

For a vector $\mathbf{x} = (x_1, x_2, \cdots, x_n)^T$, its **Euclidean norm** is defined as

$$\|\mathbf{x}\| = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}.$$

Example. If $\mathbf{x} = (1, 3, 1)$, we have

$$\|\mathbf{x}\| = \sqrt{1^2 + 3^2 + 1^2} = \sqrt{11}$$

Remark. We can check that the Euclidean norm satisfies all the following requirements:

1. $\|\mathbf{x}\| \geq 0$, and $\|\mathbf{x}\| = 0$ if and only if $\mathbf{x} = \mathbf{0}$, where $\mathbf{0}$ is the zero vector.
2. $\|\alpha\mathbf{x}\| = |\alpha| \cdot \|\mathbf{x}\|$, for all $\alpha \in \mathbb{R}$.
3. (Triangle Inequality): $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$.

With these conditions, we can extend the idea of Euclidean norm and define more general p-norms.

Vector norm

For a vector $\mathbf{x} = (x_1, x_2, \cdots, x_n)^T$, its ***p*-norm** is defined as

$$\|\mathbf{x}\|_p = \left(|x_1|^p + |x_2|^p + \cdots + |x_n|^p\right)^{\frac{1}{p}}, \quad 1 \leq p \leq \infty.$$

Example. For the vector $\mathbf{x} = (2, -3)^T$, $\|\mathbf{x}\|_3 = \left(|2|^3 + |-3|^3\right)^{1/3} = \sqrt[3]{35}$.

Remark. $\|\mathbf{x}\|_2$ is nothing but the Euclidean norm. Also, we can see that

$$\|\mathbf{x}\|_1 = \sum_{i=1}^{n} |x_i|, \qquad \|\mathbf{x}\|_\infty = \max_{1 \leq i \leq n} |x_i|.$$

All p-norms satisfy the three requirements of a norm:

1. $\|\mathbf{x}\|_p \geq 0$, and $\|\mathbf{x}\|_p = 0$ if and only if $\mathbf{x} = \mathbf{0}$, where $\mathbf{0}$ is the zero vector.
2. $\|\alpha\mathbf{x}\|_p = |\alpha| \cdot \|\mathbf{x}\|_p$, for all $\alpha \in \mathbb{R}$.
3. (Triangle Inequality): $\|\mathbf{x} + \mathbf{y}\|_p \leq \|\mathbf{x}\|_p + \|\mathbf{y}\|_p$.

Calculus

Let's also have a review on calculus

- Rate of change of functions
- Differentiation
- Integration
- Optimization

Limit of functions

Definitions:

- Left-hand limit: We say that $\lim_{x \to a^-} f(x) = L$ if $f(x)$ is close enough to L whenever x is close enough to a and $x < a$.
- Right-hand limit: We say that $\lim_{x \to a^+} f(x) = L$ if $f(x)$ is close enough to L whenever x is close enough to a and $x > a$.
- Two-sided limit: We say that $\lim_{x \to a} f(x) = L$ if both the left-hand limit and the right-hand limit exist and are equal, i.e. $\boxed{\lim_{x \to a} f(x) = L \iff \lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x) = L}$

Remark: Whether f is defined at a or the value of f at a is **NOT** important for finding $\lim_{x \to a^-} f(x)$, $\lim_{x \to a^+} f(x)$, $\lim_{x \to a} f(x)$. Example: If $f(x) = \begin{cases} -x & \text{if } x < 0 \\ 1 & \text{if } x = 0 \\ x^2 & \text{if } x > 0 \end{cases}$, we have

$\lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} (-x) = 0$ and $\lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} x^2 = 0$,

so the two-sided limit exists and we have $\lim_{x \to 0} f(x) = 0$ $(\neq 1)$

Properties of limits of functions

If $\lim_{x\to a} f(x)$ and $\lim_{x\to a} g(x)$ exist, then

- $\lim_{x\to a} f(x) \pm g(x) = \lim_{x\to a} f(x) \pm \lim_{x\to a} g(x)$
- $\lim_{x\to a} cf(x) = c \lim_{x\to a} f(x)$ (where c is a constant)
- $\lim_{x\to a} f(x)g(x) = \left(\lim_{x\to a} f(x)\right) \cdot \left(\lim_{x\to a} g(x)\right)$
- $\lim_{x\to a} \dfrac{f(x)}{g(x)} = \dfrac{\lim_{x\to a} f(x)}{\lim_{x\to a} g(x)}$ (if $\lim_{x\to a} g(x) \neq 0$)

Examples:

- $\lim_{x\to 0} \left(\dfrac{1}{x} - \dfrac{1}{x^2+x}\right) = \lim_{x\to 0} \dfrac{(x+1)-1}{x(x+1)} = \lim_{x\to 0} \dfrac{1}{x+1} = 1$
- $\lim_{x\to 2} \dfrac{2-x}{3-\sqrt{x^2+5}} = \lim_{x\to 2} \left(\dfrac{2-x}{3-\sqrt{x^2+5}} \cdot \dfrac{3+\sqrt{x^2+5}}{3+\sqrt{x^2+5}}\right)$
  $= \lim_{x\to 2} \dfrac{(2-x)(3+\sqrt{x^2+5})}{4-x^2} = \lim_{x\to 2} \dfrac{3+\sqrt{x^2+5}}{2+x} = \dfrac{6}{4} = \dfrac{3}{2}$

Properties of limits of functions

Some other useful limit results:

- $\lim\limits_{x\to 0} \dfrac{e^x - 1}{x} = 1$
- $\lim\limits_{x\to 0} \dfrac{\ln(1+x)}{x} = 1$
- $\lim\limits_{x\to 0} \dfrac{\sin x}{x} = 1$

Examples:

- $\lim\limits_{x\to 0} \dfrac{e^{3x} - 1}{x} = \lim\limits_{x\to 0} \dfrac{e^{3x} - 1}{3x} \cdot 3 = 1 \cdot 3 = 3$
- $\lim\limits_{x\to 0} \dfrac{\sin 2x}{\sin 3x} = \lim\limits_{x\to 0} \dfrac{\frac{\sin 2x}{2x}(2x)}{\frac{\sin 3x}{3x}(3x)} = \dfrac{\left(\lim\limits_{x\to 0} \dfrac{\sin 2x}{2x}\right) \cdot 2}{\left(\lim\limits_{x\to 0} \dfrac{\sin 3x}{3x}\right) \cdot 3} = \dfrac{1 \cdot 2}{1 \cdot 3} = \dfrac{2}{3}$

Limits at infinity

Definitions:

- We say that $\lim_{x\to\infty} f(x) = L$ if $f(x)$ is close enough to L whenever x is large enough.
- (Similar for $\lim_{x\to-\infty} f(x)$)

Examples:

- $\lim_{x\to\infty} \dfrac{1}{x-1} = 0$
- $e = \lim_{n\to\infty} \left(1+\dfrac{1}{n}\right)^n = \lim_{x\to\infty} \left(1+\dfrac{1}{x}\right)^x$
- $\lim_{x\to\infty} \left(1+\dfrac{1}{2x}\right)^{3x} = \lim_{x\to\infty} \left(1+\dfrac{1}{2x}\right)^{3x\cdot\frac{2}{2}} = \lim_{x\to\infty} \left(1+\dfrac{1}{2x}\right)^{2x\cdot\frac{3}{2}} = \left(\lim_{x\to\infty} \left(1+\dfrac{1}{2x}\right)^{2x}\right)^{\frac{3}{2}} = e^{\frac{3}{2}}$
- $\lim_{x\to\infty} \dfrac{x^k}{e^x} = 0$ and $\lim_{x\to\infty} \dfrac{(\ln x)^k}{x} = 0$ for any positive integer k

Asymptotes

- Horizontal asymptotes:
  If $\lim_{x\to\infty} f(x) = b$ **or** $\lim_{x\to-\infty} f(x) = b$,
  then $y = b$ is a horizontal asymptote of $f(x)$.
- Vertical asymptotes:
  If $\lim_{x\to a^-} f(x) = \pm\infty$ **or** $\lim_{x\to a^+} f(x) = \pm\infty$,
  then $x = a$ is a vertical asymptote of $f(x)$.

Examples:

- $y = 0$ is a horizontal asymptote of $f(x) = e^x$ since $\lim_{x\to-\infty} e^x = 0$
- $x = 2$ is a vertical asymptote of $f(x) = 1 + \dfrac{1}{x-2}$ since $\lim_{x\to 2^+} f(x) = \infty$

Derivatives

f is said to be differentiable at $x = a$ if the following limit (called the derivative of f at $x = a$) exists:

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$

Another form:

$$f'(a) = \lim_{h \to 0} \frac{f(a + h) - f(a)}{h}$$

Remark: For piecewise functions, we need to check both $\lim\limits_{x \to a^-} \dfrac{f(x) - f(a)}{x - a}$ and $\lim\limits_{x \to a^+} \dfrac{f(x) - f(a)}{x - a}$

Example of finding derivative by definition (i.e. **first principle**):

- If $f(x) = x^2$, then

$$\begin{aligned} f'(x) &= \lim_{h \to 0} \frac{f(x + h) - f(x)}{h} \\ &= \lim_{h \to 0} \frac{(x + h)^2 - x^2}{h} = \lim_{h \to 0} \frac{2xh + h^2}{h} = 2x. \end{aligned}$$

Derivatives of common functions

- $(x^n)' = nx^{n-1}$
- $(e^x)' = e^x$
- $(\ln x)' = \dfrac{1}{x}$
- $(\sin x)' = \cos x$
- $(\cos x)' = -\sin x$
- $(\tan x)' = \sec^2 x = \dfrac{1}{\cos^2 x}$
- $(c)' = 0$ (where c is a constant)
- $(\sinh x)' = \cosh x$ (where $\sinh x = \dfrac{e^x - e^{-x}}{2}$, $\cosh x = \dfrac{e^x + e^{-x}}{2}$)
- $(\cosh x)' = \sinh x$
- $(\tanh x)' = \operatorname{sech}^2 x = \dfrac{1}{\cosh^2 x}$

Differentiation rules

If f and g are differentiable at a point, then the following functions are also differentiable at that point:

- $(f(x) \pm g(x))' = f'(x) \pm g'(x)$
- $(cf(x))' = cf'(x)$ (where c is a constant)
- Product rule:
$$(fg)'(x) = f'(x)g(x) + f(x)g'(x)$$
- Quotient rule:
$$\left(\frac{f}{g}\right)'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2} \quad (\text{if } g(x) \neq 0)$$

Examples:

- $(x^3 \sin x)' = (x^3)' \sin x + x^3 (\sin x)' = 3x^2 \sin x + x^3 \cos x$
- $\left(\frac{\sin x}{x^2+1}\right)' = \frac{(\sin x)'(x^2+1) + (\sin x)(x^2+1)'}{(x^2+1)^2} = \frac{(x^2+1)\cos x + 2x \sin x}{(x^2+1)^2}$

Differentiation rules

Chain rule:

If $f(x)$ is differentiable at $x = a$ and $g(u)$ is differentiable at $u = f(a)$, then $(g \circ f)$ is differentiable at $x = a$ and we have

$$(g \circ f)'(a) = g'(f(a))f'(a)$$

In other words, we have

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

Examples:

- $(\sin x^2)' = \frac{d(\sin u)}{du}\frac{du}{dx}$ (let $u = x^2$) $= (\cos u)(2x) = 2x \cos x^2$
- $(e^{\sin x})' = e^{\sin x} \cos x$

A more complicated version: $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx}$

Example:

- $(\ln(\cos(x^3)))' = \frac{1}{\cos x^3} \cdot (-\sin(x^3)) \cdot (3x^2) = -3x^2 \tan x^3$

Derivatives of some other special functions

- **More general exponential function**:
  Let $a > 0$ and define $a^x = e^{x \ln a}$. Then we have:
  - $a^{x+y} = a^x \cdot a^y$ for any $x, y \in \mathbb{R}$
  - $\lim\limits_{x \to 0} \dfrac{a^x - 1}{x} = \ln a$
  - $(a^x)' = a^x \ln a$

  Example: $\left(2^{x^2+\cos x}\right)' = \left(2^{x^2+\cos x} \ln 2\right)(2x - \sin x)$
- **Inverse functions**:
  If $f(y)$ is a bijective and differentiable function with $f'(y) \neq 0$ for any y, then the inverse function $y = f^{-1}(x)$ is differentiable:

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$$

  Examples:

$$(\sin^{-1} x)' = \frac{1}{\sqrt{1-x^2}}, \quad (\cos^{-1} x)' = -\frac{1}{\sqrt{1-x^2}}, \quad (\tan^{-1} x)' = \frac{1}{1+x^2}$$

Higher order derivatives

- **Second derivative**:
$$y'' = f''(x) = \frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right)$$
- ***n*-th derivative**:
$$y^{(n)} = f^{(n)}(x) = \frac{d^ny}{dx^n} = \frac{d}{dx}\left(\frac{d}{dx}\left(\frac{d}{dx}\left(\cdots\frac{dy}{dx}\right)\right)\right)$$
- 0-**th derivative**:
$$y^{(0)} = f^{(0)}(x) = f(x)$$

Example:
$(\sin x^2)'' = ((\sin x^2)')' = ((\cos x^2)(2x))' = (-\sin x^2)(2x)(2x) + 2\cos x^2 = -4x^2 \sin x^2 + 2\cos x^2$

Local extrema, critical points, turning points

Local maximum: $f(x)$ has a local maximum at $x = a$ if $f(x) \leq f(a)$ for all x near a (more precisely, for all $x \in D \cap (a - \delta, a + \delta)$ where D is the domain and $\delta > 0$ is some small number).
Local minimum: $f(x)$ has a local minimum at $x = a$ if $f(x) \geq f(a)$ for all x near a.
Note: Local extremum points can be either interior points or endpoints!
Example: For $f : [-\pi, \pi] \to \mathbb{R}$ with $f(x) = \sin x$,
local maximum points $= (-\pi, 0), (\frac{\pi}{2}, 1)$
local minimum points $= (-\frac{\pi}{2}, -1), (\pi, 0)$.
Critical points:
f has a critical point at $x = a$ if $f'(a) = 0$ or $f'(a)$ does not exist.
Turning points:
f has a turning point at $x = a$ if f' changes sign at a.
Note: $\{\text{Turning points}\} \subset \{\text{Critical points}\}$
Example: $x = 0$ is a critical point of $f(x) = x^3$, but it is not a turning point.

First and second derivative tests

Theorem:
Let $f(x)$ be a continuous function. If $f(x)$ has a local maximum/ minimum at $x = a$, then $x = a$ must be a critical point of $f(x)$.

First derivative test:

> Let $f(x)$ be a continuous function and $x = a$ be a critical point.
> (i) If f' changes sign from $+$ to $-$ at a, then $f(x)$ has a local maximum at $x = a$.
> (ii) If f' changes sign from $-$ to $+$ at a, then $f(x)$ has a local minimum at $x = a$.

Second derivative test:

> Let $f(x)$ be a continuous function.
> (i) If $f'(a) = 0$ and $f''(a) < 0$, then $f(x)$ has a local maximum at $x = a$.
> (ii) If $f'(a) = 0$ and $f''(a) > 0$, then $f(x)$ has a local minimum at $x = a$.

Increasing/decreasing functions and derivatives

- f is (monotonic) increasing on (a, b) (i.e. $f(x) \leq f(y)$ for all $x, y \in (a, b)$ with $x < y$) if and only if $f'(x) \geq 0$ on (a, b).
- f is (monotonic) decreasing on (a, b) (i.e. $f(x) \geq f(y)$ for all $x, y \in (a, b)$ with $x < y$) if and only if $f'(x) \leq 0$ on (a, b).
- f is constant on (a, b) if and only if $f'(x) = 0$ on (a, b).
- f is strictly increasing on (a, b) (i.e. $f(x) < f(y)$ for all $x, y \in (a, b)$ with $x < y$) **if** $f'(x) > 0$ on (a, b).
- f is strictly decreasing on (a, b) (i.e. $f(x) > f(y)$ for all $x, y \in (a, b)$ with $x < y$) **if** $f'(x) < 0$ on (a, b).

Indefinite integration

Indefinite integral:

- Let $f(x)$ be continuous. An antiderivative (or primitive function) of $f(x)$ is a function $F(x)$ such that $F'(x) = f(x)$.
- The collection of all antiderivatives of $f(x)$ is called the indefinite integral of $f(x)$ and is denoted by $\int f(x)dx$.
- We have $\int f(x)dx = F(x) + C$, where C is a constant.

Example: x^2, $x^2 + 3$, $x^2 - 1$ are antiderivatives of $2x$. More generally, we have $\int 2x\ dx = x^2 + C$.

Properties:

- $\int (f(x) \pm g(x))dx = \int f(x)dx \pm \int g(x)dx$
- $\int kf(x)dx = k \int f(x)dx$

Example: $\int (x^3 + 2x - 1)\ dx = \frac{x^4}{4} + x^2 - x + C$

Some basic integrals

- $\int k \; dx = kx + C$
- $\int x^n \; dx = \dfrac{x^{n+1}}{n+1} + C \text{ (where } n \neq -1)$
- $\int e^x \; dx = e^x + C$
- $\int \dfrac{1}{x} \; dx = \ln|x| + C$
- $\int a^x \; dx = \dfrac{1}{\ln a} a^x + C$
- $\int \sin x \; dx = -\cos x + C$
- $\int \cos x \; dx = \sin x + C$
- $\int \tan x \; dx = -\ln|\cos x| + C$
- $\int \sec x \; dx = \ln|\sec x + \tan x| + C$
- $\int \sec^2 x \; dx = \tan x + C$
- $\int \sec x \tan x \; dx = \sec x + C$
- $\int \csc x \cot x \; dx = -\csc x + C$
- $\int \dfrac{1}{\sqrt{1-x^2}} \; dx = \sin^{-1} x + C$
- $\int \dfrac{1}{1+x^2} \; dx = \tan^{-1} x + C$

Some other integration techniques

Integration by substitution:

$$\boxed{\int f(u(x))\frac{du}{dx}dx = \int f(u)du}$$

Example: $\int \sqrt{3x+4}\ dx = ?$

Solution: Let $u = 3x + 4$, then $\frac{du}{dx} = 3 \Rightarrow du = 3dx$. We have

$$\int \sqrt{3x+4}\ dx = \int \sqrt{u} \cdot \frac{1}{3}du = \frac{2}{9}u^{3/2} + C = \frac{2}{9}(3x+4)^{3/2} + C$$

Integration by parts:

$$\boxed{\int u\ dv = uv - \int v\ du}$$

Example:

$$\int \ln x\ dx = x\ln x - \int x\ d(\ln x) = x\ln x - \int x \cdot \frac{1}{x}\ dx = x\ln x - x + C$$

Definite integration

Definite integral: Let $a \leq b$ and $f(x)$ be a continuous function on $[a, b]$. The definite integral $\int_a^b f(x)\ dx$ is defined as the signed area under the graph of $y = f(x)$ between $x = a$ and $x = b$.

Fundamental theorem of calculus:

Suppose $F'(x) = f(x)$ on $[a, b]$. Then $\int_a^b f(x)\ dx = F(b) - F(a)$.

Example:

$$\int_1^2 x^2\ dx = \left[\frac{x^3}{3}\right]_1^2 = \frac{2^3}{3} - \frac{1^3}{3} = \frac{7}{3}$$

Definite integration

More properties of definite integrals:

- $\int_a^a f(x)\ dx = 0$
- $\int_a^b kf(x)\ dx = k\int_a^b f(x)\ dx$
- $\int_a^b (f(x) \pm g(x))\ dx = \int_a^b f(x)\ dx \pm \int_a^b g(x)\ dx$
- $\int_a^b f(x)\ dx = \int_a^c f(x)\ dx + \int_c^b f(x)\ dx \quad$ if $c \in (a, b)$

Coming back to our AI tasks

More mathematically, the data points are usually expressed in the following form:

$$\mathcal{D} = \{(\mathbf{X}_i, y_i) : i = 1, \ldots, n\}.$$

Each input $\mathbf{X}_i$ (can be 1-dimensional or high-dimensional) has a corresponding target y_i.

- A target can be a measured quantity or an annotated category.
- The model should predict the target for a new input.

The above-mentioned mathematical and statistical tools can be applied to all three tasks!

- Regression
- Classification
- Generative modelling

Example: Linear Regression

Given a set of 2D data points $\{(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)\}$, where the x-part is the explanatory variable and the y-part is the dependent variable.

Goal: To predict the y values using the x values.

To achieve this, we consider "fitting" a line through the data:

$$\boxed{y = a + bx}$$

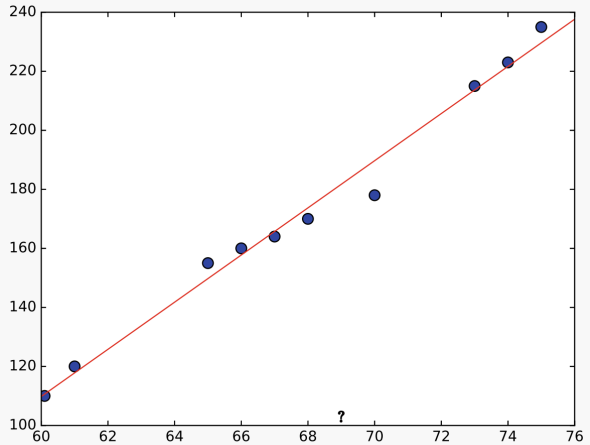
where a (the y-intercept) and b (the slope) are parameters.

Example

Example: Fitting a line to height and weight

Consider the following data set that describes a set of heights and weights.

height (in)	weight (lbs)
66	160
68	170
60	110
70	178
65	155
61	120
74	223
73	215
75	235
67	164
69	?



(Source: Mathematical Foundations for Data Analysis)

We can easily predict the weight with a specific height after fitting a line. But how to fit a suitable line?

Residual

- Let $f(x) = a + bx$.
- For every value x, we can get a predicted value $\hat{y} = f(x)$.
- Therefore, for every data point (x_i, y_i), we can get a predicted value with x_i:

$$\hat{y}_i = f(x_i) = a + bx_i$$

- We can then examine how close $\hat{y}_i$ is to y_i for each x_i. The difference is called a **residual**:

$$r_i = y_i - \hat{y}_i = y_i - f(x_i)$$

- Note that r_i can be viewed as the signed distance from y_i to the corresponding point on the straight line with the same x value.

Residual sum of squares (RSS)

Now, we want to:

- measure the overall error for the entire dataset
- avoid the effect of the sign of individual r_i's on the overall error

This motivates us to consider the **residual sum of squares (RSS)**, also known as the **sum of squared errors (SSE)** or the **sum of squared residuals (SSR)**:

$$RSS = \sum_{i=1}^{n} r_i^2 = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = \sum_{i=1}^{n} (y_i - f(x_i))^2 = \sum_{i=1}^{n} (y_i - a - bx_i)^2$$

We can then solve for the line $y = a + bx$ that minimizes the RSS to get the "best-fit" line.

Method 1: Solving for the best-fit line via linear algebra

Note that we can rewrite the problem as a matrix problem:

$$\begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

Finding the least-squares solution to $A\mathbf{x} = \mathbf{b}$ is equivalent to solving the normal equation $A^T A\mathbf{x} = A^T\mathbf{b}$. Therefore, we have

$$\boxed{\begin{bmatrix} a \\ b \end{bmatrix} = (M^T M)^{-1} M^T \mathbf{y}}$$

where $M = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$.

Method 1: Solving for the best-fit line via linear algebra

More explicitly, we have

$$M^T M = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ x_1 & x_2 & \cdots & x_n \end{bmatrix} \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} = \begin{bmatrix} n & \sum_{i=1}^{n} x_i \\ \sum_{i=1}^{n} x_i & \sum_{i=1}^{n} x_i^2 \end{bmatrix}$$

and

$$M^T \mathbf{y} = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ x_1 & x_2 & \cdots & x_n \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^{n} y_i \\ \sum_{i=1}^{n} x_i y_i \end{bmatrix}.$$

Hence, we have $\boxed{\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{bmatrix}^{-1} \begin{bmatrix} \sum y_i \\ \sum x_i y_i \end{bmatrix}}$.

Method 2: Solving for the best-fit line via calculus

Let $R = \sum_{i=1}^{n}(y_i - a - bx_i)^2$.

To find a and b that minimize R, we take partial derivatives of R with respect to a and b (remark: just treat them as usual differentiation!) and equate them to 0

$$\frac{\partial R}{\partial a} = 0 \Rightarrow \frac{\partial}{\partial a}\sum_{i=1}^{n}(y_i - a - bx_i)^2 = 0$$

$$\Rightarrow 2\sum_{i=1}^{n}(y_i - a - bx_i)(-1) = 0 \Rightarrow \sum_{i=1}^{n} y_i = na + b\sum_{i=1}^{n} x_i$$

Similarly, we have

$$\frac{\partial R}{\partial b} = 0 \ \Rightarrow \frac{\partial}{\partial b}\sum_{i=1}^{n}(y_i - a - bx_i)^2 = 0 \Rightarrow \sum_{i=1}^{n} x_i y_i = a\sum_{i=1}^{n} x_i + b\sum_{i=1}^{n} x_i^2$$

Method 2: Solving for the best-fit line via calculus

Putting the two equations together, we have

$$\begin{cases} na + b\sum_{i=1}^{n} x_i = \sum_{i=1}^{n} y_i \\ a\sum_{i=1}^{n} x_i + b\sum_{i=1}^{n} x_i^2 = \sum_{i=1}^{n} x_i y_i \end{cases}$$

Hence, we have

$$\begin{bmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum x_i y_i \end{bmatrix} \Rightarrow \boxed{\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{bmatrix}^{-1} \begin{bmatrix} \sum y_i \\ \sum x_i y_i \end{bmatrix}}$$

Same result as Method 1!

Nonlinear Regression

In fact, we can suitably extend the above methods for other regression models:

- Quadratic Regression:
$$y = a + bx + cx^2$$
- Cubic Regression:
$$y = a + bx + cx^2 + dx^3$$
- Polynomial Regression:
$$y = c_0 + c_1x + c_2x^2 + \cdots + c_nx^n$$
- Power Regression
$$y = ax^b$$
- Exponential Regression:
$$y = ab^x$$
- Logarithmic Regression:
$$y = a + b\log(x)$$

Next time

More on Linear Algebra:

- Scalars, vectors, matrices and tensors
- Dot products, cosine similarity
- L1/L2 norms, distances, orthogonality
- Applications: similarity, embeddings